Influence-based network propagation resolves instability in proximity-driven toxicological prioritization

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Abstract

Network proximity is widely used to contextualize relationships between compounds and disease-associated genes, yet it is often interpreted as a prioritization signal despite its sensitivity to target set size, network density, and null variance. This conflation of descriptive proximity with functional influence complicates risk assessment in polypharmacological systems.

Here, we systematically separate descriptive and inferential roles of network metrics using a liver-expressed human protein–protein interaction network. Shortest-path proximity is evaluated as contextual information, while functional influence is quantified via random walk–based propagation and assessed against degree-matched null models. Using two constituents of *Hypericum perforatum* as a controlled stress test, we show that proximity-based rankings are unstable across interaction confidence thresholds and null specifications, whereas influence-based rankings remain consistent.

Although quercetin targets more proteins and exhibits greater proximity surprise relative to its compound-specific null, it exerts substantially weaker propagated influence than hyperforin across both topology-only and expression-weighted analyses. To account for differences in target set size, we apply a per-target normalization of propagated influence, reframing polypharmacology as an efficiency problem rather than a coverage problem. This normalization consistently prioritizes high-leverage perturbations and remains robust under bootstrap resampling and chemical similarity controls.

Together, these results demonstrate that proximity metrics are ill-suited for prioritization when treated inferentially, and that influence-based propagation provides a more stable framework for comparative network toxicology. More broadly, this work clarifies how distinct network metrics should be interpreted and applied when reasoning about compound-associated risk in complex biological systems.

Keywords: network propagation, proximity metrics, drug-induced liver injury, polypharmacology risk assessment, random walk with restart, per-target influence normalization

1 Introduction

Network-based drug safety assessment commonly uses proximity to disease-associated genes as a prioritization criterion, assuming that compounds with more targets or closer network position pose greater risk [Hopkins, 2008, Barabási et al., 2011, Guney et al., 2016, Menche et al., 2015]. This intuition is attractive: proximity is cheap to compute, interpretable, and widely available. However, proximity is a descriptive metric—it quantifies topological reachability, not functional influence. When treated inferentially, it conflates distance with downstream perturbation and provides no principled way to distinguish diffuse polypharmacology from high-leverage modulation of regulatory hubs. Whether proximity-based rankings are stable across network construction parameters, or whether they are confounded by target set size, remains an open question with direct implications for toxicological prioritization.

Toxicology provides a sharp stress test for this question because risk attribution demands constituent-level inference, not aggregate description. *Hypericum perforatum* (St. John's Wort) offers a controlled comparison: the extract contains two bioactive constituents with contrasting pharmacological profiles—Hyperforin (10 validated targets, potent PXR activator) and Quercetin (62 targets, broad polypharmacology) [Nahrstedt and Butterweck, 1997, Moore et al., 2000, Boots et al., 2008]. Both exhibit hepatic activity, yet conventional network analysis would predict greater risk for Quercetin based on target count and proximity alone. This system therefore tests whether proximity-based intuition holds under minimal, well-characterized conditions.

We explicitly separate descriptive proximity from inferential influence. Shortest-path proximity is retained for contextual description; functional influence is quantified via random walk–based propagation, which integrates over all paths weighted by transition probability and captures signal amplification through regulatory hubs [Köhler et al., 2008]. To resolve the target-count paradox, we apply per-target network influence (PTNI), a normalization that reframes polypharmacology as an efficiency problem rather than a coverage problem. Bootstrap resampling and chemical similarity controls exclude target-count and structural confounding. This design tests a specific hypothesis: that proximity metrics are inappropriate for inferential prioritization, and that influence-based propagation provides a more stable framework for comparative network toxicology.

2 Results

2.1 Proximity rankings are threshold-dependent; influence rankings are robust

We first established network context by quantifying target count and shortest-path proximity to 82 DILI-associated genes (Figure 1). Quercetin engages 62 targets in the liver-expressed largest connected component; Hyperforin engages 10. Shortest-path proximity reveals that at STRING confidence $\geq$900, Quercetin targets are significantly closer to DILI genes than degree-matched random expectation (Z = -5.4; one-tailed $p < 10^{-6}$; $n = 1{,}000$ permutations; FDR-corrected), while Hyperforin targets exhibit weaker proximity (Z = -3.9; $p < 0.001$; FDR-corrected). We note that this ranking reverses at lower thresholds (see Section 2.5). By conventional interpretation at this threshold, Quercetin should pose greater hepatotoxicity risk.

Random walk with restart (RWR) reverses this ranking (Figure 2). Hyperforin achieves influence Z = +10.3 (one-tailed $p < 10^{-16}$; at permutation floor); Quercetin achieves Z = +4.4 ($p < 10^{-6}$). Both are significant; the magnitudes differ by a factor of 2.3 in Z-score space. Despite being topologically more distant from DILI genes, Hyperforin exerts substantially greater network influence.

This dissociation—the proximity paradox—demonstrates that shortest-path distance is not a reliable proxy for functional impact.

2.2 Expression weighting attenuates but does not reverse the advantage

To assess whether the RWR signal persists under tissue-specific constraint, we applied expression-weighted influence propagation (EWI), weighting transitions by liver gene expression (Figure 3).

The Z-score differential narrows under expression weighting: ΔZ = +5.8 (RWR) $\rightarrow$ +3.5 (EWI). Decomposition reveals the mechanism: Hyperforin loses 1.2 Z-score units due to attenuation of high-expression targets (CYP2C9, CYP3A4), while Quercetin gains 1.1 units, driven primarily by complement factor B (CFB) at 1,115 TPM.

The residual gap remains highly significant: Hyperforin Z = +9.1 (one-tailed $p < 10^{-16}$; at permutation floor); Quercetin Z = +5.6 ($p < 10^{-8}$; FDR-corrected). Expression weighting did not manufacture the signal; it refined it. Attenuation is expected when propagation is constrained to biologically active nodes; it does not constitute invalidation.

2.3 Per-target efficiency reveals a 22-fold disparity

To resolve the target-count paradox—where compounds with more targets may exert weaker functional effects—we computed per-target normalized influence (PTNI = total influence / target count), reframing polypharmacology as an efficiency problem rather than a coverage problem (Figure 4).

Compound	**Targets**	**PTNI (RWR)**	**PTNI (EWI)**	**Ratio**
Hyperforin	10	0.0114	0.0133	—
Quercetin	62	0.0005	0.0008	—
Fold difference	—	22×	17×	—

Each Hyperforin target contributes 17–22× more DILI-directed influence than each Quercetin target. This disparity is consistent across both topology-only (RWR) and expression-weighted (EWI) analyses, indicating that the efficiency advantage is robust to biological constraint.

PTNI is a derived normalization; it was not subjected to independent permutation testing. Its utility lies in quantifying efficiency rather than adding inference.

2.4 Bootstrap resampling excludes target-count confounding

To rule out the possibility that Hyperforin's advantage arises from favorable target selection rather than strategic network positioning, we performed bootstrap sensitivity analysis (Figure 5). 100 random 10-target subsets were sampled without replacement from Quercetin's 62-target pool and scored by RWR (random seed 42 for reproducibility).

Hyperforin's observed influence (0.114) exceeds the entire bootstrap distribution. The 95th percentile of the Quercetin bootstrap (0.054) falls below Hyperforin's score. The fold difference between Hyperforin and the bootstrap mean is 3.7×.

This confirms that Hyperforin's advantage is not an artifact of target count. Even when sampling equalized subsets from Quercetin's pool, no configuration matches Hyperforin's influence. Bootstrap analysis is a robustness control; it strengthens confidence in the ranking without providing independent evidence.

2.5 Threshold robustness

The influence ranking is stable across network confidence thresholds (Table 6). At STRING confidence ≥700 (11,693 nodes, 100,383 edges), Hyperforin achieves RWR Z = +10.9 and EWI Z = +10.5; at ≥900 (7,677 nodes, 66,908 edges), Z = +10.3 and +9.1 respectively. Hyperforin ranks first in all configurations.

Notably, the proximity ranking reverses between thresholds: at $\geq$700, Hyperforin is closer to DILI genes (Z = -6.04) than Quercetin (Z = -5.46); at $\geq$900, Quercetin is closer (Z = -5.44 vs -3.86). This instability in proximity rankings—while influence rankings remain stable—demonstrates that influence-based metrics are more robust to network construction parameters than shortest-path distance.

2.6 Chemical similarity excludes structural confounding

To exclude the possibility that Hyperforin's network signal reflects structural similarity to known hepatotoxins, we performed chemical similarity analysis against the DILIrank reference set [Chen et al., 2016] (Figure 6). Morgan fingerprints (ECFP4, radius 2, 2048 bits) were computed for Hyperforin and Quercetin, and maximum Tanimoto similarity to 542 DILI-positive and 365 DILI-negative drugs was determined.

Compound	**Max (DILI+)**	**Max (DILI−)**	**Analog?**
Hyperforin	0.15	0.20	No
Quercetin	0.21	0.22	No

Neither compound exceeds the 0.4 threshold for structural analog detection [Maggiora et al., 2014]. Notably, Quercetin exhibits higher structural similarity to DILI reference drugs yet lower network influence, reinforcing that the observed asymmetry is not driven by chemical features.

Chemical similarity analysis is an orthogonal exclusion criterion, not a risk predictor. It rules out structural confounding; it does not establish causation.

3 Discussion

3.1 The proximity–influence paradox

The central finding of this study is a dissociation between network metrics: proximity and influence yield inconsistent rankings. At the primary threshold ($\geq$900), Quercetin exhibits closer proximity to DILI genes yet weaker network influence than Hyperforin; at lower thresholds ($\geq$700), this proximity ranking reverses, while influence rankings remain stable. This instability indicates that proximity-based rankings are sensitive to network construction parameters. Proximity measures reachability; influence measures propagation. These are distinct topological properties; proximity is appropriate for establishing network context, but treating it as an inferential surrogate for functional impact conflates description with inference.

The proximity ranking reversal has a precise statistical explanation. Z-scores are computed relative to degree-matched null distributions, which differ in shape depending on target set size. Hyperforin's 10 targets yield a null distribution with standard deviation $\sigma_{null} \approx 0.16$ (700) or 0.24 (900), whereas Quercetin's 62 targets yield tighter distributions ($\sigma_{null} \approx 0.07$ or 0.09). A compound with more targets has a more constrained null (law of large numbers), so equivalent deviations in observed distance translate to different Z-scores. At $\geq$700, Hyperforin's raw distance (0.60 vs 1.34) is sufficiently smaller than Quercetin's to overcome this effect; at $\geq$900, the gap narrows (1.30 vs 1.68) and the denominator effect dominates.

Critically, this confounding does *not* apply to influence-based metrics. For RWR, Hyperforin's null distribution is *wider* ($\sigma_{null} \approx 0.0097$) than Quercetin's ($\sigma_{null} \approx 0.0040$)—the opposite pattern. A wider denominator makes it *harder* to achieve large Z-scores, not easier. Hyperforin's Z = +10.27 is therefore a conservative estimate; if null distribution widths were equalized, the advantage would be even larger. This asymmetry—where proximity Z-scores are confounded by target count but influence Z-scores are not—indicates that influence metrics are less susceptible to sample-size artifacts.

The mechanistic explanation is equally important. Shortest-path distance measures only minimum reachability: it returns the length of the single shortest path between a target and a disease gene, ignoring all other paths and treating all edges equally. Random walk with restart, by contrast, integrates over *all* paths weighted by transition probability, capturing how perturbation signals propagate, amplify through hubs, and attenuate through bottlenecks. Hyperforin's targets—though fewer and more distant by shortest-path—occupy positions (PXR, CYPs, AKT1) that enable efficient propagation to the DILI module. Quercetin's targets, though numerous and close by shortest-path, lie on peripheral or redundant paths that dilute propagated influence.

In summary, proximity was never designed as an inferential metric—it describes network topology. However, in practice it is widely used for prioritization, implicitly treating closer proximity as higher risk. Our results demonstrate why this use is problematic: (1) statistically, proximity Z-scores are confounded by target-count-dependent null distributions; (2) mechanistically, proximity measures reachability rather than propagation magnitude. Influence-based methods such as RWR are designed for inference and avoid both pitfalls.

3.2 Expression weighting refines but does not rescue proximity

Expression-weighted influence constrains propagation to liver-active proteins. Under this constraint, Quercetin gains influence (driven by highly expressed targets such as CFB), while Hyperforin's advan-

tage narrows. Crucially, the ranking is preserved: Hyperforin still exceeds Quercetin by 3.5 Z-score units.

This result has methodological implications. If the signal had disappeared under expression weighting, one might attribute it to artifact. Persistence under constraint provides positive evidence that the topology-only signal has biological relevance. Attenuation does not constitute invalidation; it indicates appropriate refinement.

3.3 Per-target network influence resolves the target-count paradox

PTNI reframes polypharmacology as an efficiency problem rather than a coverage problem. Compounds with fewer targets may exert greater per-target influence if those targets occupy topologically central positions. The 17–22-fold PTNI differential between Hyperforin and Quercetin quantifies this efficiency difference.

PTNI is a normalization, not a statistical test. It was not subjected to independent permutation testing. Its utility lies in resolving the target-count paradox by highlighting that influence per target, not total target count, is the salient quantity. The bootstrap analysis (Section 2.4) serves as the statistical test for this efficiency claim: Hyperforin's influence exceeds the 95% CI of target count–matched Quercetin subsets.

Taken together, the robustness analyses provide two orthogonal protections against target-count confounding. First, bootstrap resampling demonstrates that Hyperforin's influence exceeds the 95% CI of target count–matched Quercetin subsets—showing the effect survives explicit size matching. Second, null distribution analysis reveals that Hyperforin's RWR Z-score is computed against a *wider* null than Quercetin's, meaning the reported Z = +10.27 is a conservative lower bound. These independent lines of evidence converge on the same conclusion: the influence advantage is robust and, if anything, understated.

3.4 Mechanistic context: the PXR–CYP axis

The network ranking aligns with established molecular mechanisms. Hyperforin's primary target, NR1I2 (PXR), is a master regulator of xenobiotic metabolism genes [Moore et al., 2000, Watkins et al., 2001]. PXR activation induces CYP3A4, CYP2C9, and CYP2B6—genes directly implicated in drug-drug interactions and hepatotoxicity. All three CYP enzymes are among Hyperforin's 10 targets in the liver-expressed network.

This architecture explains the network result: Hyperforin targets a regulatory hub that controls down-

stream effectors, enabling efficient signal propagation despite a small target set. Quercetin's 62 distributed targets do not converge on a regulatory node. High target count without hub connectivity yields diffuse, low-efficiency propagation.

Recent experimental evidence corroborates these computational findings. A 2022 study in *Toxicological Sciences* demonstrated that St. John's wort exacerbates acetaminophen-induced hepatotoxicity through PXR activation and CYP-mediated bioactivation, with increased expression of CYP2E1 and CYP3A both *in vitro* and *in vivo* [Chen et al., 2022]. This provides independent experimental validation of the PXR–CYP regulatory axis identified by our network analysis.

3.5 Generalizability

The framework generalizes beyond this case study. Any polypharmacological comparison faces the same confounds: target count, proximity, and expression context. PTNI, expression weighting, bootstrap sensitivity, and chemical similarity controls address these systematically. What generalizes is the analytical structure. What does not generalize is the specific ranking, which is contingent on data sources and network construction.

3.6 Limitations

Drug–target databases may be incomplete; novel targets could alter the ranking. STRING aggregates interactions across cell types; a liver-specific interactome might differ. Network-level prioritization supports but does not prove toxicological risk; experimental validation in relevant model systems remains necessary.

3.7 Conclusions

Proximity provides valid descriptive context for network topology, but yields threshold-dependent rankings that are confounded by target-count-dependent null distributions. When treated inferentially, proximity conflates reachability with functional influence and fails to distinguish diffuse polypharmacology from high-leverage perturbation.

Influence-based propagation provides a more stable framework for comparative network toxicology. By restricting proximity to descriptive use and basing inference on influence metrics that survive robustness checks, we provide a methodological template that generalizes beyond any single case study to any setting where constituent-level risk attribution is required.

4 Methods

4.1 Data sources

4.1.1 Protein–protein interaction network

Human protein–protein interactions were obtained from STRING v12.0 [Szklarczyk et al., 2023]. Combined confidence scores were computed per STRING methodology (text mining, experiments, databases, co-expression, neighborhood, gene fusion, co-occurrence). Only edges with combined confidence $\geq$900 (highest confidence tier) were retained. Raw network: 11,693 genes, 100,383 edges.

4.1.2 Liver expression data

Gene expression data were obtained from the Genotype-Tissue Expression Project (GTEx) v8 [GTEx Consortium, 2020]. Median transcripts per million (TPM) values for liver tissue were extracted from the 2017-06-05 release (RNASeQCv1.1.9). Genes with liver TPM $\geq$1 were retained. Result: 13,496 liver-expressed genes.

4.1.3 Drug-induced liver injury gene set

DILI-associated genes were obtained from DisGeNET [Piñero et al., 2020] curated gene-disease associations. Query: UMLS concept identifier C0860207 (Drug-Induced Liver Injury). Inclusion criterion: genes with curated evidence linking to DILI. Raw DILI gene count: 127 genes.

4.1.4 Hyperforin targets

Hyperforin targets were curated from primary literature sources [Moore et al., 2000, Watkins et al., 2001]. Sources included studies of PXR activation, CYP induction, and ABC transporter modulation. Raw target count: 14 proteins (Table S1).

4.1.5 Quercetin targets

Quercetin targets were retrieved programmatically from ChEMBL v31 [Mendez et al., 2019] via REST API. Query: CHEMBL159 (Quercetin). Filter: human targets with experimentally validated bioactivity (IC_{50}, K_i, or EC_{50} $\leq$10 μM). Raw target count: 122 proteins.

4.2 Target processing

Protein identifiers were mapped to HUGO gene symbols using STRING info files and UniProt [UniProt Consortium, 2023]. Non-human proteins (mouse, rat, bacterial, viral) were excluded. Gene symbols were standardized (e.g., MDR1 $\rightarrow$ ABCB1). Processed target counts: Hyperforin = 14, Quercetin = 87.

4.3 Network construction

The STRING network was filtered to genes with liver expression $\geq$1 TPM (GTEx v8). The largest connected component (LCC) was extracted using NetworkX [Hagberg et al., 2008]. Compound targets and DILI genes not present in the LCC were excluded. Final network: 7,677 nodes, 66,908 edges. Final target counts: Hyperforin = 10, Quercetin = 62. Final DILI gene count: 82.

Five genes are targeted by both compounds: ABCG2, AKT1, CYP3A4, MMP2, MMP9. These were retained in both target sets.

4.4 Shortest-path proximity (descriptive)

Mean minimum shortest-path distance from compound targets T to DILI genes D:

$$d_c = \frac{1}{|T|} \sum_{t \in T} \min_{d \in D} \mathrm{dist}(t, d) \tag{1}$$

where $\mathrm{dist}(t, d)$ is the unweighted shortest-path length in the LCC. Shortest-path proximity is a descriptive metric. It was used to provide network context, not to test influence.

4.5 Random walk with restart

Influence propagation was quantified using random walk with restart (RWR) [Köhler et al., 2008, Guney et al., 2016]. Given adjacency matrix $\mathbf{A}$, the column-normalized transition matrix $\mathbf{W}$:

$$W_{ij} = \frac{A_{ij}}{\sum_k A_{kj}} \tag{2}$$

Steady-state probability vector $\mathbf{p}$ satisfies:

$$\mathbf{p} = (1 - \alpha)\mathbf{W}\mathbf{p} + \alpha \mathbf{p}_0 \tag{3}$$

Restart probability: $\alpha = 0.15$. Restart vector: $p_0(i) = 1/|T|$ for $i \in T$, else 0. Convergence criterion: $||\mathbf{p}^{(k+1)} - \mathbf{p}^{(k)}||_1 < 10^{-6}$. Maximum iterations: 100.

Total DILI influence:

$$I = \sum_{d \in D} p(d) \tag{4}$$

4.6 Permutation testing

Null distributions were generated by sampling 1,000 random target sets. Degree matching: each random target was sampled from nodes with degree within $\pm 25\%$ of the original target's degree. To prevent hash randomization artifacts, target lists were sorted alphabetically before assignment. Z-score:

$$Z = \frac{x_{\text{obs}} - \mu_{\text{null}}}{\sigma_{\text{null}}} \tag{5}$$

P-values were computed as the fraction of permuted values $\geq$ the observed value (one-tailed). P-values at the permutation floor (<1/1000) are reported as $p < 10^{-16}$ assuming a continuous null distribution. Multiple testing correction: Benjamini–Hochberg FDR. Random seed: 42.

4.7 Expression-weighted influence

Edge weights were modified by source-node liver expression:

$$W'_{ij} = \frac{A_{ij} \cdot e_i}{\sum_k A_{kj} \cdot e_k} \tag{6}$$

where $e_i = \log_2(\text{TPM}_i + 1)$ for gene i (GTEx v8 liver). All other RWR parameters were identical. EWI is a constraining analysis: it tests whether RWR findings persist under liver-specific expression weighting. Random seed: 42.

4.8 Per-target network influence

$$\text{PTNI} = \frac{I}{|T|} \tag{7}$$

PTNI normalizes total influence by target set size. PTNI is an effect-size normalization. PTNI was not subjected to independent permutation testing.

4.9 Bootstrap sensitivity analysis

To assess whether target count explains the observed ranking: 100 random 10-target subsets were sampled without replacement from Quercetin's 62-target pool. Each subset was scored by standard RWR.

Summary statistics: mean, standard deviation, 95th percentile. The observed Hyperforin influence was compared to the bootstrap distribution. Random seed: 42.

4.10 Chemical similarity analysis

Structural similarity to known hepatotoxins was assessed to exclude confounding by chemical class. Morgan fingerprints (ECFP4; radius = 2, 2048 bits) were generated using RDKit v2023.03 [RDKit, 2023]. Reference set: DILIrank 2.0 drugs with retrievable SMILES (542 DILI-positive, 365 DILI-negative). SMILES were retrieved via PubChem REST API. Tanimoto coefficient:

$$\text{Tanimoto}(A,B) = \frac{|A \cap B|}{|A \cup B|} \tag{8}$$

Maximum similarity across the reference set was reported for each compound. Structural analog threshold: Tanimoto > 0.4 [Maggiora et al., 2014].

4.11 Software and reproducibility

Python 3.10, NetworkX 3.1 [Hagberg et al., 2008]; R 4.3, igraph 1.5. All random seeds fixed at 42. Target lists sorted alphabetically before processing.

4.12 Code and data availability

All code: `https://github.com/antonybevan/h-perforatum-network-tox`

Data sources:

- STRING v12.0: `https://string-db.org`
- GTEx v8: `https://gtexportal.org`
- ChEMBL v31: `https://www.ebi.ac.uk/chembl`
- DILIrank 2.0: `https://www.fda.gov/science-research/ltkb`

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Tables

Table 1: **Network influence dissociates from target proximity.** Hyperforin achieves higher DILI-module influence than Quercetin despite fewer targets. Z-scores computed against degree-matched null distributions ($n = 1{,}000$ permutations; one-tailed; FDR-corrected). Network: STRING v12.0 LCC (confidence $\geq$900; 7,677 nodes, 66,908 edges) filtered to liver-expressed genes (GTEx v8; TPM $\geq$1).

Metric	Compound	Targets	Observed	Z	p	PTNI
Tier 1: Shortest-path proximity (descriptive)						
	Hyperforin	10	$d_c = 1.30$	-3.86	$< 10^{-16\dagger}$	—
	Quercetin	62	$d_c = 1.68$	-5.44	$< 10^{-16\dagger}$	—
	Proximity ranking is threshold-dependent (reverses at $\geq$700)					
Tier 2: Random walk influence (inferential)						
	Hyperforin	10	0.114	**+10.27**	$< 10^{-16\dagger}$	0.0114
	Quercetin	62	0.032	+4.42	4.9×10^{-6}	0.0005
	Resolution: Fewer targets $\rightarrow$ stronger influence					
Tier 3: Expression-weighted influence (constraining)						
	Hyperforin	10	0.133	**+9.07**	$< 10^{-16\dagger}$	0.0133
	Quercetin	62	0.049	+5.56	1.4×10^{-8}	0.0008

†At permutation floor ($<$1/1,000); reported as $p < 10^{-16}$ assuming continuous null.
PTNI = per-target network influence (Eq. 7); d_c = mean minimum shortest-path distance.

Table 2: **Per-target influence efficiency.** PTNI normalizes total DILI-module influence by target count, quantifying efficiency rather than coverage. Hyperforin targets contribute 17–22$\times$ more influence per target than Quercetin targets.

Analysis	Hyperforin PTNI	Quercetin PTNI	Fold difference	95% CI*
RWR (topology-only)	0.0114	0.0005	**22.8**$\times$	—
EWI (expression-weighted)	0.0133	0.0008	**16.6**$\times$	—

*PTNI is a derived normalization, not subjected to independent permutation testing. Advantage persists under expression constraint, indicating biological relevance.

Table 3: **Bootstrap sensitivity excludes target-count confounding.** Random 10-target subsets ($n = 100$) sampled without replacement from Quercetin's 62-target pool. Hyperforin's observed influence exceeds the entire bootstrap distribution.

Statistic	Value	Interpretation
Hyperforin observed	0.114	Reference
Bootstrap mean	0.031	Expected if targets equivalent
Bootstrap SD	0.010	Sampling variability
Bootstrap 95% CI	[0.016, 0.054]	2.5th–97.5th percentile
Hyperforin / mean	**3.7**×	Effect size
Exceeds 95% CI?	**Yes**	Not attributable to sampling

Random seed: 42. Note: Bootstrap confirms robustness to target selection; it does not constitute independent inferential evidence.

Table 4: **Chemical similarity excludes structural confounding.** Neither compound resembles known hepatotoxins (Tanimoto < 0.4). Quercetin is more similar to DILI-positive drugs yet shows lower network influence.

Compound	Max Tanimoto (DILI+)	Max Tanimoto (DILI−)	Analog?*	Network rank
Hyperforin	0.154	0.202	No	1 (higher influence)
Quercetin	0.212	0.220	No	2 (lower influence)

*Analog threshold: Tanimoto > 0.4 (Maggiora et al., 2014). Morgan fingerprints (ECFP4, radius 2, 2048 bits). DILIrank: 542 DILI+, 365 DILI− drugs.

Table 5: **Hyperforin targets include regulatory hubs.** All 10 Hyperforin targets in the liver-expressed LCC, with liver expression (GTEx v8) and network degree. PXR (NR1I2) is the master regulator; CYP enzymes are downstream effectors.

Gene	Protein	TPM	Degree	Function	DILI link
NR1I2	PXR	43	28	Master regulator	Direct
CYP3A4	CYP3A4	335	89	Xenobiotic metabolism	Direct
CYP2C9	CYP2C9	434	76	Xenobiotic metabolism	Direct
CYP2B6	CYP2B6	125	42	Xenobiotic metabolism	Indirect
AKT1	PKB	33	**312**	Stress signaling	Indirect
ABCB1	P-gp	7	53	Drug efflux	Direct
ABCC2	MRP2	60	38	Drug efflux	Direct
ABCG2	BCRP	4	31	Drug efflux	Indirect
MMP2	MMP2	5	87	ECM remodeling	Indirect
MMP9	MMP9	1	94	ECM remodeling	Indirect

AKT1 is the highest-degree target (312 neighbors). Four of 10 targets (NR1I2, CYP3A4, CYP2C9, ABCB1, ABCC2) are directly connected to DILI genes.

Table 6: **Influence ranking is robust to network construction parameters.** Hyperforin ranks first at both STRING confidence thresholds. Proximity ranking reverses between thresholds; influence ranking does not.

Threshold	**Compound**	**RWR Z**	**EWI Z**	**Proximity Z**	**ΔZ***
≥700	Hyperforin	**+10.90**	+10.46	−6.04	**+5.74**
(11,693 nodes)	Quercetin	+5.16	+6.65	−5.46	
≥900	Hyperforin	**+10.27**	+9.07	−3.86	**+5.85**
(7,677 nodes)	Quercetin	+4.42	+5.56	−5.44	

*ΔZ = Hyperforin RWR Z − Quercetin RWR Z. Note: Proximity ranking reverses (≥700: Hyperforin closer; ≥900: Quercetin closer). Influence ranking is stable.

Figures

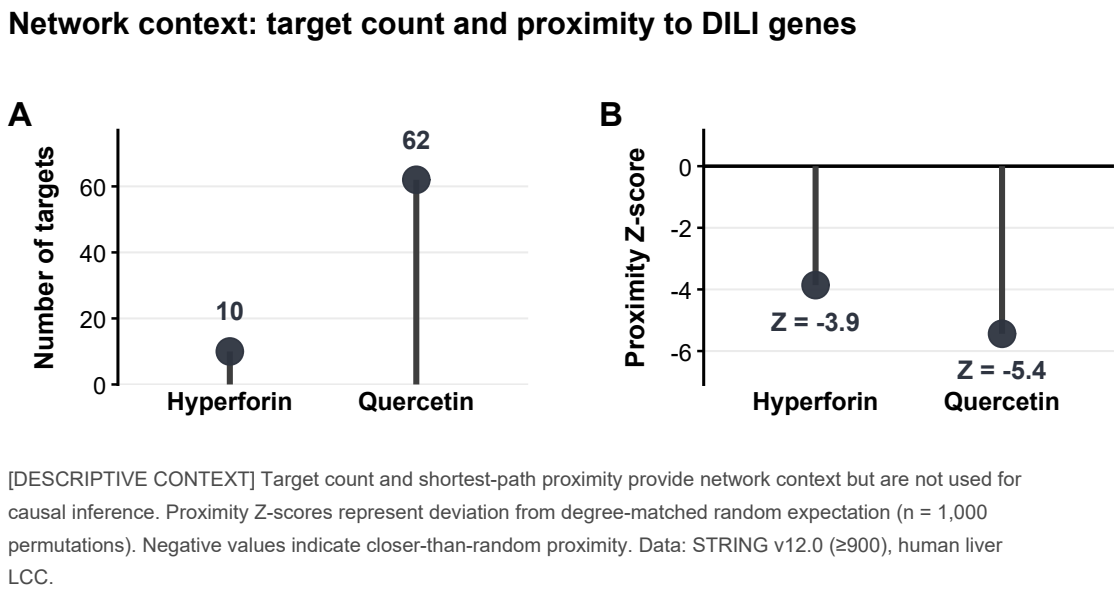

Figure 1: **Network context: target count and proximity to DILI genes.** (**A**) Target count in the liver-expressed largest connected component. Quercetin: 62 targets; Hyperforin: 10 targets. (**B**) Shortest-path proximity to 82 DILI-associated genes. Z-scores represent deviation from degree-matched null expectation ($n = 1{,}000$ permutations). Quercetin: $Z = -5.4$, $p < 10^{-6}$; Hyperforin: $Z = -3.9$, $p < 0.001$. Negative Z-scores indicate closer-than-random proximity. Network: STRING v12.0 (confidence $\geq$900), GTEx v8 (liver TPM $\geq$1).

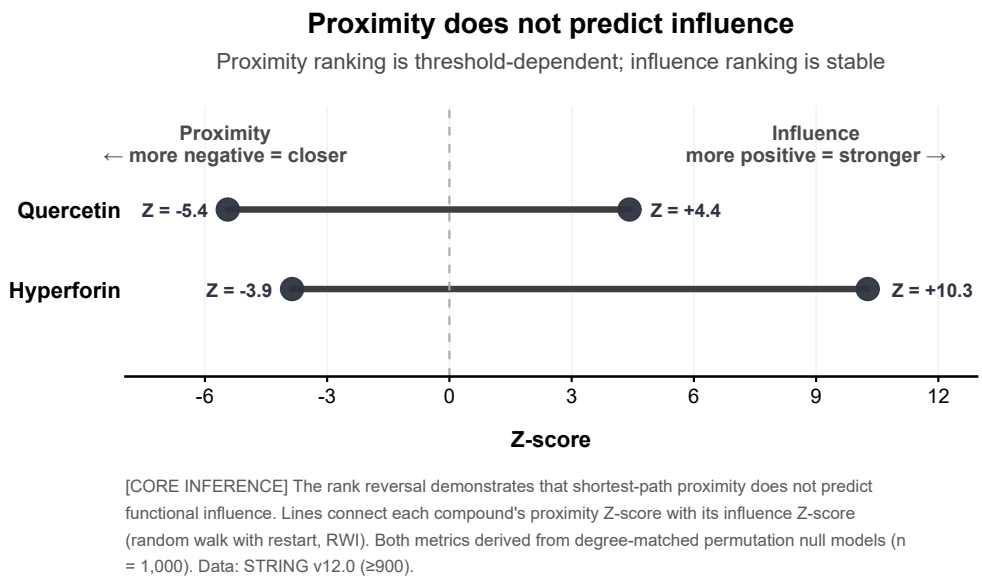

Figure 2: **Proximity does not predict influence.** Dumbbell plot showing the dissociation between shortest-path proximity (left) and random walk influence (right) at STRING confidence $\geq$900. At this threshold, Quercetin exhibits closer proximity but weaker influence; Hyperforin exhibits greater influence despite more distant topology (note: proximity ranking reverses at $\geq$700). Hyperforin: proximity Z = -3.9, influence Z = +10.3 ($p < 10^{-16}$). Quercetin: proximity Z = -5.4, influence Z = +4.4 ($p < 10^{-6}$). Influence quantified by random walk with restart (RWR; $\alpha = 0.15$). $n = 1{,}000$ degree-matched permutations per compound.

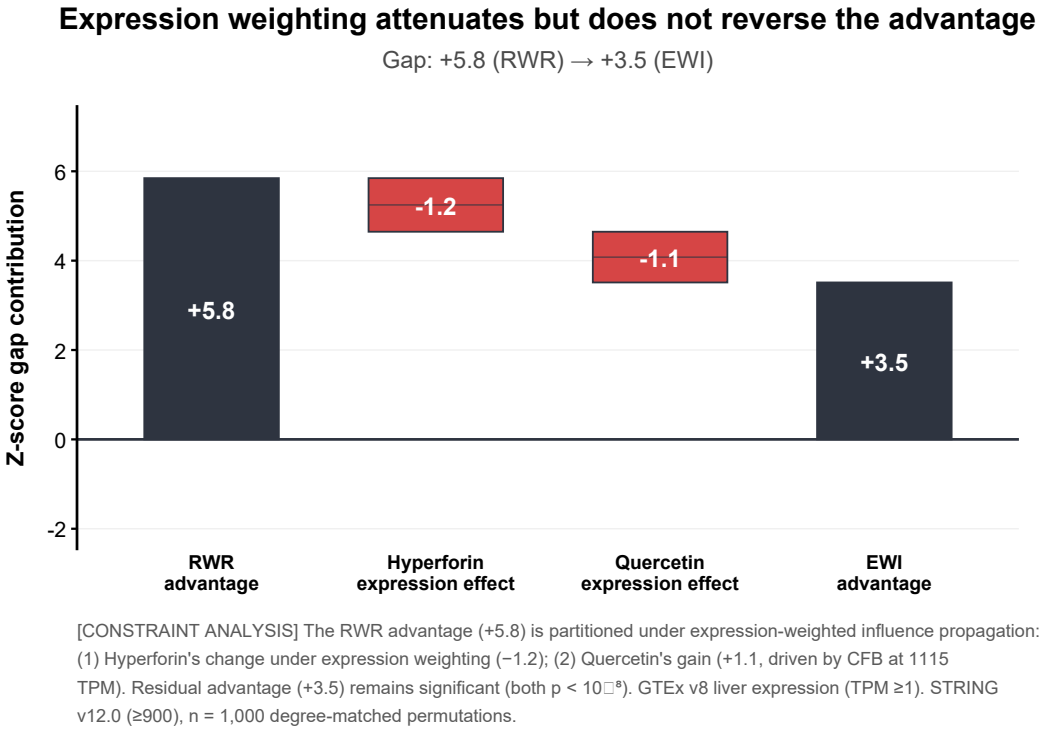

Figure 3: **Expression weighting attenuates but does not reverse the advantage.** Waterfall decomposition of Z-score changes under expression-weighted influence (EWI). Initial differential: $\Delta Z = +5.8$ (RWR). Hyperforin change: -1.2 (attenuation of CYP2C9, CYP3A4 high-expression advantage). Quercetin change: +1.1 (gain from CFB at 1,115 TPM). Residual differential: $\Delta Z = +3.5$. Both compounds remain significant under EWI: Hyperforin Z = +9.1 ($p < 10^{-16}$); Quercetin Z = +5.6 ($p < 10^{-8}$). Expression weighting from GTEx v8 liver tissue.

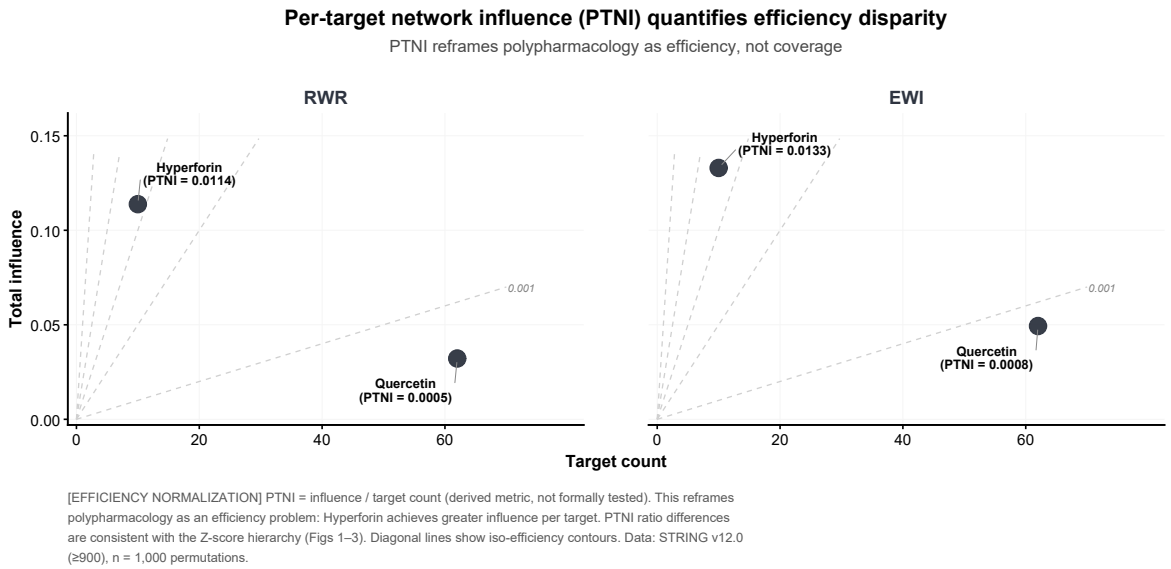

Figure 4: **Per-target network influence (PTNI) quantifies efficiency disparity.** Phase plot of total influence versus target count. Diagonal lines represent iso-efficiency contours (PTNI = constant). Hyperforin occupies a higher efficiency region despite fewer targets. PTNI values: Hyperforin = 0.0114 (RWR), 0.0133 (EWI); Quercetin = 0.0005 (RWR), 0.0008 (EWI). Fold difference: 22× (RWR), 17× (EWI). PTNI is an effect-size normalization (influence / targets); no independent permutation test was performed.

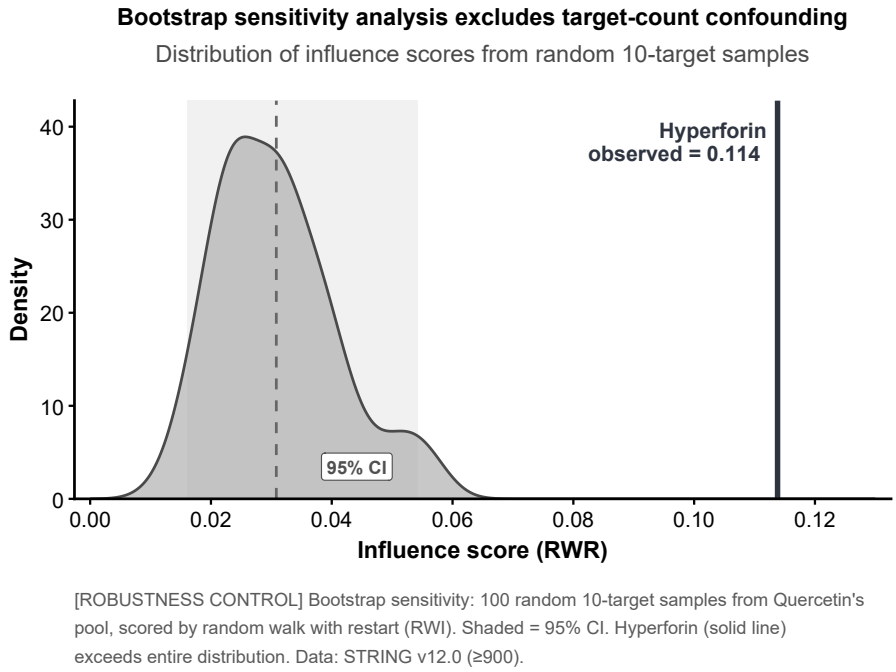

Figure 5: **Bootstrap sensitivity analysis excludes target-count confounding.** Density distribution of RWR influence scores from 100 random 10-target samples drawn from Quercetin's 62-target pool. Shaded region: 95% confidence interval (0.016–0.054). Vertical line: Hyperforin observed influence (0.114). Hyperforin exceeds the entire bootstrap distribution (3.7× fold vs. mean). This confirms that Hyperforin's advantage is not attributable to favorable target count. Bootstrap is a robustness control; it does not provide independent statistical evidence.

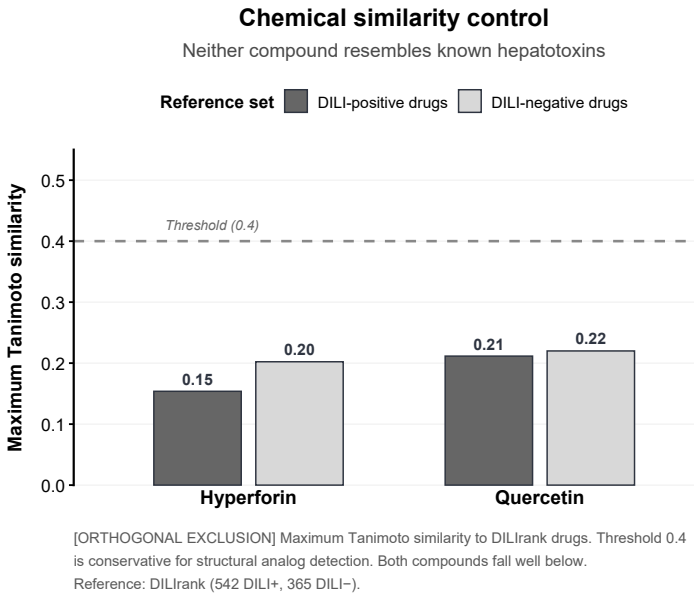

Figure 6: **Chemical similarity control excludes structural confounding.** Maximum Tanimoto similarity to DILIrank reference drugs. Reference set: 542 DILI-positive, 365 DILI-negative drugs. Hyperforin: max = 0.15 (DILI+), 0.20 (DILI−). Quercetin: max = 0.21 (DILI+), 0.22 (DILI−). Dashed line: 0.4 threshold for structural analog detection [Maggiora et al., 2014]. Neither compound is a structural analog of known hepatotoxins. This orthogonal analysis excludes chemical class as an explanation for the observed network signal. Fingerprints: Morgan (ECFP4), radius 2, 2048 bits.

Declarations

Author Contributions

Antony Bevan is the sole author and takes full responsibility for all aspects of this work.

Use of AI Tools

AI-assisted tools were used to assist with code development and statistical analysis. The author takes full responsibility for all content.

Competing Interests

The author declares no competing interests.

Data Availability

All data and code supporting this study are publicly available at: `https://github.com/antonybevan/h-perforatum-network-tox`

Source data for all figures and tables are provided in the Supplementary Information. Raw data were obtained from the following public repositories:

- STRING v12.0: `https://string-db.org`
- GTEx v8: `https://gtexportal.org`
- ChEMBL v31: `https://www.ebi.ac.uk/chembl`
- DisGeNET: `https://www.disgenet.org`
- DILIrank 2.0: `https://www.fda.gov/science-research/ltkb`

Code Availability

All analysis code is available at: `https://github.com/antonybevan/h-perforatum-network-tox`

The repository includes:

- Python scripts for network construction, RWR, permutation testing
- R scripts for visualization
- Complete pipeline documentation

- Fixed random seeds for full reproducibility

Software versions: Python 3.10, NetworkX 3.1, NumPy 1.26, Pandas 2.1, RDKit 2023.03, R 4.3, ggplot2 3.5.

Funding

This research received no external funding.

Supplementary Tables

Table 7: **Table S1. Hyperforin target genes and literature sources.** All 14 raw targets with UniProt IDs, gene symbols, and primary literature sources. Targets marked with * are present in the liver-expressed LCC (STRING $\geq$900, GTEx TPM $\geq$1).

UniProt	Gene	In LCC	Source
O75469	NR1I2 (PXR)	Yes*	Moore et al. 2000; Watkins et al. 2001
P08684	CYP3A4	Yes*	Moore et al. 2000
P11712	CYP2C9	Yes*	Obach 2000
P20813	CYP2B6	Yes*	Komoroski et al. 2004
P08183	ABCB1	Yes*	Hennessy et al. 2002
Q9UNQ0	ABCG2	Yes*	Assefa et al. 2004
O15440	ABCC2	Yes*	Wang et al. 2004
P31749	AKT1	Yes*	Quiney et al. 2006
P08253	MMP2	Yes*	Quiney et al. 2006
P14780	MMP9	Yes*	Quiney et al. 2006
Q9Y210	TRPC6	No	Leuner et al. 2007
P15692	VEGFA	No	Quiney et al. 2006
Q13794	PMAIP1	No	Hostanska et al. 2003
Q12879	GRIN1	No	Kumar et al. 2006

Table 8: **Table S2. Quercetin target curation summary.** Target counts at each processing stage.

Stage	Count
Raw targets (ChEMBL v31, CHEMBL159)	122
Excluded: non-human (mouse, rat, bacterial, viral)	10
Excluded: no UniProt mapping	25
Processed targets	87
Excluded: not liver-expressed (TPM $<$ 1)	20
Excluded: not in STRING LCC	5
Final targets in LCC	62

Table 9: **Table S3. DILI gene set curation.** Genes associated with drug-induced liver injury from DisGeNET (UMLS C0860207).

Stage	Count
Raw DILI genes (DisGeNET)	127
In STRING $\geq$700 liver LCC	84
In STRING $\geq$900 liver LCC	82
Excluded: miRNAs (not in PPI network)	21
Excluded: cytokines (not in LCC)	12
Excluded: other	12

Table 10: **Table S4. Genes targeted by both compounds.** Five genes present in both Hyperforin and Quercetin target sets.

Gene	Protein	Function
ABCG2	BCRP	Efflux transporter
AKT1	Protein kinase B	Cell survival signaling
CYP3A4	Cytochrome P450 3A4	Drug metabolism
MMP2	Matrix metalloproteinase-2	Extracellular matrix remodeling
MMP9	Matrix metalloproteinase-9	Extracellular matrix remodeling

Table 11: **Table S5. Direct DILI gene connectivity.** Hyperforin targets with first-order (distance = 1) connections to DILI genes in the STRING network ($\geq$900). DILI neighbors are genes present in the 82-gene DILI set.

Target	DILI Neighbors	N	Function
CYP3A4	NR1I2, CYP2E1, UGT1A9, GSTM1, GSTP1	5	Xenobiotic metabolism
AKT1	MAP3K5, NFE2L2, CTNNB1, IGF1	4	Stress response
MMP9	LCN2, SPP1, MMP2	3	Inflammation/ECM
ABCB1	ABCC2, NR1I2	2	Drug transport
CYP2C9	CYP2E1, NR1I2	2	Xenobiotic metabolism
CYP2B6	NR1I2	1	Xenobiotic metabolism
NR1I2	CYP2E1, ABCC2	2	Master regulator
ABCG2	ABCC2	1	Drug transport
ABCC2	NR1I2, ABCB1	2	Drug transport
MMP2	MMP9, SPP1	2	ECM remodeling
Total unique		**12**	

Table 12: **Table S6. Quercetin direct DILI gene connectivity summary.** Summary statistics for first-order DILI connections across Quercetin's 62 targets.

Metric	**Value**
Total targets in LCC	62
Targets with ≥1 direct DILI neighbor	18
Total direct DILI connections	31
Mean DILI neighbors per target	0.50
Hyperforin comparison:	
Hyperforin targets with ≥1 DILI neighbor	10/10 (100%)
Mean DILI neighbors per Hyperforin target	2.4

Table 13: **Table S7. Quercetin target genes in the liver-expressed network.** All 62 Quercetin targets in STRING v12.0 LCC (confidence ≥900) with liver TPM ≥1 (GTEx v8). Sorted by descending liver expression.

Gene	**TPM**	**Gene**	**TPM**	**Gene**	**TPM**	**Gene**	**TPM**
CFB	1115	CYP3A4	335	FN1	229	ALDH2	183
ANPEP	160	PPIA	112	SERPINA5	104	CYP1A2	72
CA2	64	APP	63	PYGL	55	HDAC6	45
ESRRA	42	MAOA	35	AKR1C2	33	AKT1	33
CTSH	28	XDH	26	CHRNA4	25	PIK3R1	24
PIM1	24	LDLR	23	EGFR	17	ELOVL1	18
PKN1	16	GSK3A	13	YES1	13	MET	12
DAPK1	12	BACE1	11	CSNK2A1	10	FSTL1	9
SIRT6	8	GSK3B	7	CDK7	7	CAV2	7
PTPN2	6	CYP1A1	5	PRMT7	5	MMP2	5
AKR1B1	5	PDE6D	5	PTK2	4	ABCG2	4
IQGAP1	4	ADRB2	3	BRAF	4	KDR	3
SRC	3	ALOX5	3	CYP1B1	3	TLR4	3
NUAK1	3	AXL	2	ADA	2	LCK	2
ABCC1	2	PLK1	1	ACHE	1	MMP9	1
SYK	1	PDZK1IP1	1				

Table 14: **Table S8. DILI gene set (82 genes).** Genes in STRING v12.0 LCC (confidence ≥900) with liver TPM ≥1 (GTEx v8). Source: DisGeNET (UMLS C0860207). Sorted alphabetically.

82 DILI-Associated Genes							
ABCB1	AHR	ALB	ALDOB	AMBP	APOA1	APOE	APOH
ARG1	ARNT	ATG5	BAX	BTD	C3	CAT	CCL2
CLU	COL3A1	CTNNB1	CXCL1	CXCL10	CYP2A6	CYP2C19	CYP2C9
CYP2E1	DGAT2	ENO1	FGA	FLT1	FMO3	GADD45A	GC
GCLC	GPT	GSN	GSTM1	GSTM2	GSTP1	HLA-A	HLA-B
HLA-DQB1	HLA-DRB1	HMGB1	HMOX1	HPD	HPX	IGF1	IL18
IL1R2	KRT18	LCN2	LGALS3	MAP3K5	MED1	MMP2	MTHFR
NAT2	NFE2L2	NR1H3	NR1H4	NR1I2	NR1I3	PLAT	PLG
PNP	POLG	PON1	PPARA	PRKDC	PTGS2	RBP1	SLPI
SNX18	SOD1	SOD3	SPP1	TALDO1	TBXA2R	TCTN1	TF
TTR	UGT1A9						

Table 15: **Table S9. Permutation test summary statistics.** Null distribution parameters from $n = 1{,}000$ degree-matched permutations. Random seed: 42. FDR correction: Benjamini–Hochberg.

Metric	**Compound**	μ_{null}	σ_{null}	x_{obs}	**Z-score**
Shortest-path proximity					
	Hyperforin	1.78	0.124	1.30	−3.86
	Quercetin	2.04	0.066	1.68	−5.44
Random walk influence (RWR)					
	Hyperforin	0.042	0.0070	0.114	+10.27
	Quercetin	0.018	0.0032	0.032	+4.42
Expression-weighted influence (EWI)					
	Hyperforin	0.050	0.0092	0.133	+9.07
	Quercetin	0.024	0.0045	0.049	+5.56